

SECTION 16270-A
CAST RESIN ENCAPSULATED DRY TRANSFORMERS
(480V-22.8KV)

PART 1 - GENERAL

1.1 DESCRIPTION

This specification covers designing, furnishing, and installing and testing.

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1.2 APPLICABLE CODES AND STANDARDS

The standards and codes applicable to only a portion of the work specified in this specification are referenced in the relevant parts or clauses. Standards and codes which are generally applicable to the work of this specification are listed below:

- A. A. IEC 60076-11 Power transformers dry-type transformers
- B. CNS - Chinese national Standard 13390 C4468 Dry Type Transformers

Where references are made to standards, the Contractor may submit equipment built and tested according to other standards providing that the requirements of the referenced standards are complied with as a minimum.

The year of revision of each standard and code is given to facilitate references. Nevertheless, the revision current at the date of tender shall be the basis of the contract.

If there is any conflict between this specification and any reference standard or code, the Contractor shall inform the Owner and ask for clarification before manufacture.

1.3 SUBMITTALS

The Contractor shall submit the following for all equipment components and accessories:

- A. Data: Transformer specification sheet as similar to the appendix B of ANSI C57.12.00 - 1980.

- B. Certificates and Reports: Certification that the transformers will meet the test of IEC 60726, Section 5.
- C. If aluminum windings are used for transformers, the Contractor shall submit the formula to calculate the temperature rise and short circuit current density in order that the short circuit temperature rise of aluminum winding will be limited to 200°C.

1.4 WORK INCLUDED

- A. A. Furnish the cast resin encapsulated dry transformers.
- B. Follow local government regulations and get approval of energizing all transformers.

PART 2 - PRODUCTS

2.1 SERVICE CONDITIONS

A. Altitude

A height above sea-level not exceeding 1000m.

B. Temperature of cooling air

Never exceeding 40°C and never below 0°C. In addition, an ambient air temperature never exceeding the following values:

1. 33°C average in any one day.
2. 20°C average in any one year.

C. Wave-shape of supply voltage

A supply voltage of which the wave-shape is approximately sinusoidal.

D. Symmetry of polyphase supply voltage.

For polyphase transformers, supply voltages which are approximately symmetrical.

E. The transformers must be provided repairs serving and protecting firmly in 5 years.

2.2 DESIGN REQUIREMENTS

- A. The transformers shall be vacuum impregnated or machine roving cast-resin encapsulated, dry-type, three-phase, 60 HZ, with at least 155°C insulation for high-tension and low-tension side (130°C insulation is only permissible for aluminum strip winding on high-tension side). Transformer shall be rated for continuous operation at rated KVA with a maximum winding temperature rise of 100°C (80°C for 130°C insulation) and shall be enclosed in ventilation steel enclosures. Forced air cooling shall be provided, where shown, such that the transformer rating can be increased by at least one third without exceeding the specified temperature rise above the specified maximum ambient temperature.
- B. Internally connected no - load taps, two 2-1/2 percent above the rated voltage and two 2-1/2 percent below the rated voltage. Externally operated tap changer for deenergized operation, lifting eyes, diagrammatic nameplates, and accessories for forced air cooling shall be provided . Taps shall be accessible from front or rear of unit.
- C. Transformer impedance shall be as shown with NEMA Standard Tolerance.
- D. Transformer shall be mounted on approved vibration and noise isolating and damping supports. Core and coil assemblies shall be rigidly braced to withstand mechanical forces under line fault conditions and to resist vibration and shock forces during shipment.
- E. Each high voltage winding and each low voltage winding of each phase shall be separately cast as a rigid tubular coil of copper or aluminum conductors.
- High voltage and low voltage windings shall be coaxial with no mechanical connection. The windings shall not absorb moisture and shall be suitable for both storage and operation in adverse environments, including prolonged storage in 100% humidity at temperatures ranging from 0°C to 40°C and shall be capable of being energized immediately after such storage without predrying.
- F. The transformers shall have a delta primary - grounded wye secondary phase configuration unless noted otherwise.

The phase relationship shall be standard: primary to lead secondary by 30°.

- G. All transformer winding terminations shall be provided with NEMA CC 1 Standard copper connectors. Bus bar terminations shall be in accordance with NEMA standards and supported with insulators not more than 400 mm from the connection holes.
- H. Temperature sensing elements shall be embedded in windings.
- I. The core shall be visibly grounded to the base and the base provided with a grounding connection pad.
- J. The outside surfaces of the core shall be protected against corrosion..
- K. Coils shall be vacuum impregnated or machine roving, and cast-resin encapsulated to give it a tough moisture and dirt resistant overall finish. The coils shall be non-flammable and self-extinguishing when exposed to electric arc during fault conditions as specified by IEC Standards.
- L. The maximum sound pressure level will be 69 dbA, except, measured at 30cm from external surface at AN rating.
- M. Unless noted otherwise, the transformers shall be designed for non-liner loads with K-factor rating of K4.
- N. Transformers will be of environmental class E2 and of climatic class C2 as defined in IEC 60076
- O. These transformers will be of class F1 as defined in IEC 60076-11. F1 class will be indicated on the rating plate. This test must have been performed in accordance with IEC 60076-11.
- P. Measurement of partial discharges.
For this measurement, the acceptance criteria will be less than or equal to 5 PC at 1.30 Un.
- Q. Random wound coils are not acceptable for use in medium voltage windings.
- R. Cooling fans shall be high reliability, capable of operating 5 years continuously without failure.

- S. Transformers shall be suitable for switching by all types of switching devices, i.e. vacuum or SF6 breakers without detrimental effect on the windings
- T. The enclosure if required shall be constructed of steel and provided with openings for ventilation in accordance with IEC standard and with removable panels to facilitate cleaning and to provide access to terminals. Removable roof panels shall be provide for access to lifting eyes on the transformer frame. The base shall consist of structural steel members which will permit skidding, jacking, or rolling the assembly in any direction.

2.3 RATING

- A. Rated capacity : The capacity is planned and selected according to the equipment model selected by the manufacturer.
- B. Rated voltage / connection:
 - 1. LV 480 V, delta.
 - 2. HV 22.8 KV, wye, neutral low resistance grounded
 - 3. Connection Dyn1
- C. System insulation classes:
 - 1. LV 1.1 KV
 - 2. HV, 22.8 KV 25 KV (ANSI)
- D. Basic impulse insulation levels:
 - 1. HV, 22.8 KV 125 KV
- E. Transformers shall be suitable for impulse testing.
- F. Power frequency test voltage:
 - 1. LV 3 KV
 - 2. HV, 22.8 KV 50 KV
- G. Tap changer :

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|--------------|--|
| 1. Type | De-energized |
| b. Tap | $\pm 2 - 2 \frac{1}{2}\%$ (+5%, +2.5%, 0, -2.5%, -5%) |
| H. Impedance | 6.0% |

2.4 SHOP TESTING

A. General requirements

1. Tests shall be made at the manufacturer's works, unless otherwise agreed between the Owner and the Contractor.
2. Dielectric tests in accordance with square-source voltage withstand test, induced overvoltage withstand test and lightning impulse test shall be made with the transformer at approximately ambient temperature.
3. All external components and fittings that are likely to affect the performance of the transformer during each test shall be in place.
4. Tapped windings shall be connected on their principal tapping unless agreed by the Owner otherwise.
5. The test basis for all characteristics other than insulation is the rated condition.
6. The reference temperature to which the load loss, impedance voltage and short-circuit impedance are to be corrected shall be the maximum winding temperature rise plus 20 K. If transformers have windings of different insulation system temperatures only one reference temperature is to be used, that is, the one relating to the winding having the higher insulation system temperature.

B. Routine tests: Tests shall be performed in accordance with IEC 60076-11, and shall include as next:

1. Visual & dimensions.
2. Measurement of winding resistance.
3. Measurement of voltage ratio and check of vector relation.
4. Measurement of insulation resistance.
5. Measurement of impedance voltage and load loss.
6. Measurement of no-load loss and current.
7. Separate-source voltage withstand.
8. Induced over voltage withstand test.

9. Partial discharge measurement, after install and energize must be re-measurement.
- C. Type tests: Tests shall be performed in accordance with IEC 60076-11 and shall include the following type tests:
1. Measurement of insulation Power factor
 2. Lightning impulse test.
 3. Load temperature rise test.
 4. Sound level measurement.
- D. Sound level: The measurement of sound level shall be performed in accordance with IEC 60076-10.

2.5 SPECIAL REQUIREMENT

- A. Radio influence voltage levels shall not exceed 250 microvolt at 11.4 KV, 4.16 KV, 480 V, and 650 microvolt at 22.8 KV.

Radio influence voltage tests shall be carried out at work in accordance with NEMA TR - 1 or agreed IEC alternative.

PART 3 - EXECUTION

3.1 FIELD TEST

After the equipment has been installed, inspected, and placed in operating condition, it shall be field-tested. The field test shall demonstrate that the equipment and components function in compliance with the specification over the entire range of operation.

Required field test includes,

All standard tests recommended by factory as part of commissioning plus following tests.

A. Visual and Mechanical Inspection:

1. Check primary, secondary, and ground connections.
2. Clean and inspect bushings.
3. Inspect bushing clamps and gaskets.
4. Check accessory devices for condition and proper operation.

B. Electrical Tests:

1. Insulation Resistance Test & Polarization index test:
Megger transformer windings high to low and ground,
low to high and ground, and high and low to ground.
 2. Perform High potential test
 3. Perform turns ratio test.
 4. Where auxiliary cooling has been provided, verify
proper operation of such equipment.
- C. Test transformer to IEC standards.

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